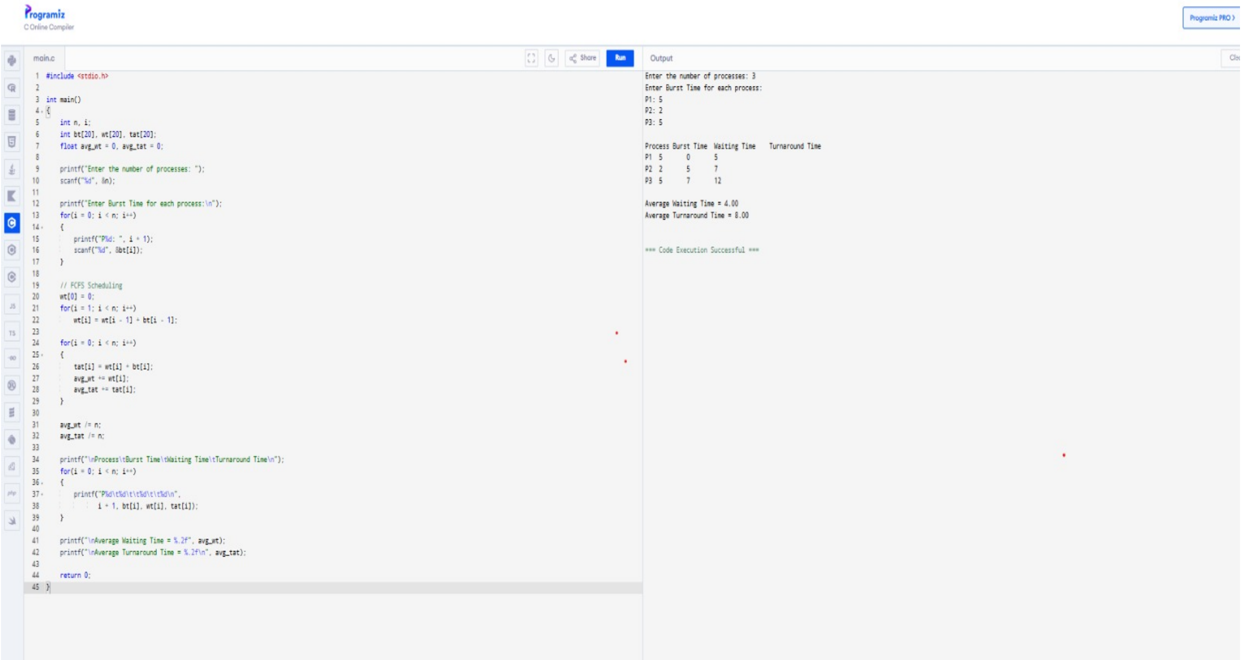
QUESTION 3:

Design a CPU scheduling program with C using First Come First Served.

CODING :



Programiz
C Online Compiler

main.c

```c
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], wt[20], tat[20];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    printf("Enter Burst Time for each process:\n");
    for(i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // FCFS Scheduling
    wt[0] = 0;
    for(i = 1; i < n; i++)
        wt[i] = wt[i - 1] + bt[i - 1];

    for(i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t\t%d\t\t%d\n",
               i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

    return 0;
}
```

Output

```
Enter the number of processes: 3
Enter Burst Time for each process:
P1: 5
P2: 2
P3: 5

Process Burst Time  Waiting Time    Turnaround Time
P1  5       0       5
P2  2       5       7
P3  5       7       12

Average Waiting Time = 4.00
Average Turnaround Time = 8.00


=== Code Execution Successful ===
```